Ruoyu Zhao

PhD Student, City University of Hong Kong | Co-founder & CTO, BrainGalaxy Ltd.

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Google Scholar | LinkedIn | GitHub

Education

PhD **City University of Hong Kong**, Department of Computer Science — *Kowloon, Hong Kong*, Sept 2025 – Present

- Supervisors: Prof. Zhichao Lu, Prof. Qingfu Zhang
- Research interests: Brain-inspired Intelligence, Neural Architecture, Optimization

B.Eng **Tsinghua University**, Department of Electronic Engineering — *Beijing, China*, Sept 2020 – Jun 2025

- Major: Electronic Information Science and Technology
- Outstanding Undergraduate Thesis Award (top 5%)

Minor **Tsinghua University**, Department of Statistics and Data Science — *Beijing, China*, Sept 2023 – Jun 2024

- Minor degree in Statistics; completed all 10 required courses within one year

Entrepreneurship

BrainGalaxy Ltd., Co-founder & Chief Technology Officer (CTO) — *Shanghai, China*, Aug 2026 – Present

Brain-inspired AI for edge and embodied intelligence: full-stack co-design of brain-inspired algorithms and memristor-based compute-in-memory chips

- Lead the company's technology direction and the algorithm R&D division (application, model architecture, deployment, and software toolchain groups).
- Drive the "algorithm first, chip next" roadmap: low-power spiking and linear-attention models for on-device LLMs and robot vision, deployed first on mature edge SoCs, then co-designed with in-house memristor compute-in-memory chips.
- Bridge research and product, turning the team's brain-inspired architecture and memristor device research into the company's core technology.

Qingguang Innovation Technology Ltd., Lead Developer — *Beijing, China*, Jan 2024 – Oct 2025

AI-assisted smart glasses for the visually impaired; co-founders: Minh David Thao Chan, Shuo Wang

- Owned the end-to-end technical roadmap, from hardware integration to on-device algorithms, on an NVIDIA Jetson Orin Nano platform.
- Built an on-device assisted-navigation pipeline combining semantic segmentation, perspective transformation, and A* path planning, adapted from autonomous-driving methods.
- Selected for Tsinghua's 3rd iStar Program (top 11 teams university-wide); third prize, 2024 China-U.S. Young Maker Competition (Beijing Division).

Publications & Manuscripts

* equal contribution, ✉ corresponding author.

[1] Neural Spectral Capacity: An Architectural Quantity from Network Specification Alone — 2026

Chenyu Zhu, ***Ruoyu Zhao***, Zhichao Lu✉

NeurIPS 2026, arXiv

[2] Fast and robust memristive cell based on 2D materials for energy-efficient in-memory computing — Accepted
Yuwan Hong, Yanming Liu, ***Ruoyu Zhao***, Zijun Wang, Zhongyuan Pang, Zhibin Wang, Yuyang Peng, Peigen Zhang, Yizhe Guo, Anhan Liu, Ao Tian, Songang Peng, Hao Guo, He Tian✉, Tian-Ling Ren
Nature Communications, accepted (*Impact Factor=18.1*)

[3] Generalizable Blind Image Quality Assessment with Retrieval Augmentation and Contrastive Vision-Text Local Meta-Training — 2026
Yan Zhong, Zhiyuan You, Xinping Zhao, Li Zhang, Xinyuan Song, ***Ruoyu Zhao***, Lei Shi, Tingting Jiang✉
ACM MM 2026

[4] Dyn-SSM: towards the efficient long sequence learning via bio-interpretable dynamics in spiking state space models — May 2026
Yan Zhong*, ***Ruoyu Zhao****, Chao Wang, Jiaqi He, Qinghui Guo, Jianguo Zhang, Zhichao Lu, Luziwei Leng✉
IEEE Transactions on Cognitive and Developmental Systems (TCDS)

[5] Energy-Efficient Collaborative Perception: A Block-Skipping DNN Approach with Dynamic Frequency Scaling and Environment Awareness — Feb 2026
Minh D. Thao Chan, ***Ruoyu Zhao***, Yukuan Jia, Zhaojun Nan, Sheng Zhou✉, Zhisheng Niu
IEEE Transactions on Intelligent Transportation Systems (T-ITS)

[6] CTD-inpainting: Towards the Coherence of Text-driven Image Inpainting in Social Media — Apr 2025
Yan Zhong, Xinping Zhao, Guangzhi Zhao, Bohua Chen, ***Ruoyu Zhao***, Fei Hao, Lei Shi✉, Li Zhang✉
Information Fusion (*Impact Factor=14.8*)

[7] Diff-2-in-1: Bridging Generation and Dense Perception with Diffusion Models — Oct 2024
Shuhong Zheng, Zhipeng Bao, ***Ruoyu Zhao***, Martial Hebert, Yu-Xiong Wang✉
ICLR 2025 (Poster)

[8] Optimization of Layer Skipping and Frequency Scaling for Convolutional Neural Networks under Latency Constraint — Jun 2024
Minh D. Thao Chan*, ***Ruoyu Zhao****, Yukuan Jia, Ruiqing Mao, Sheng Zhou✉
1st Workshop on Cooperative Intelligence for Embodied AI, ECCV 2024 (Poster)

[9] Volumetric Video Compression Through Neural-based Representation — Apr 2024
Yuang Shi, ***Ruoyu Zhao***, Simone Gasparini, Géraldine Morin, Wei Tsang Ooi✉
MMVE '24

Under review:

[10] Large Language Model-Guided Evolutionary Discovery of Native Neural Architectures for Spiking Sequence Modeling
Ruoyu Zhao*, Jiaqi Wu*, Chenyu Zhu, Zhichao Lu✉
IEEE Transactions on Evolutionary Computation (TEVC) (*Impact Factor=16.0*), arXiv

[11] Shared Gaussianization: What Gaussian Regularizers Certify About Contrastive Learning, and What They Miss
Ruoyu Zhao, Yuting Chen, Jinheng Zhang, Zhehao Zou, Tong Che✉
AISTATS 2027, arXiv

[12] Rare Gate Disagreements Can Limit Plasticity: When Gradient Flow Mispredicts Finite-Batch SGD
Ruoyu Zhao, Mingxuan Zhang, Jianbo Dai, Jiaqi Wu, Chenyu Zhu, Tong Che✉
AISTATS 2027, arXiv

[13] Neural Spectral Capacity with Residual Context for Heterogeneous Resource Allocation within Transformers
Chenyu Zhu, ***Ruoyu Zhao***, Zhichao Lu✉

ICLR 2027, arXiv

Research Experience

Tsinghua University (THU), Research Assistant — *Beijing, China*
Advisors: Prof. Yali Li and Prof. Shengjin Wang — Sept 2024 – Jun 2025

- Studied continual lifelong learning with diffusion generative models; this work became my undergraduate thesis (Outstanding Undergraduate Thesis Award, top 5%).

City University of Hong Kong (CityU), Research Assistant — *Kowloon, Hong Kong*
Advisor: Prof. Zhichao Lu — Aug 2024 – Nov 2024

- Explored efficient brain-inspired architectures and non-backpropagation learning built on the brain's sparse, local, and dynamic computation.
- Proposed a parallel computing scheme for soft-reset LIF neurons that keeps richer neural dynamics, validated on spiking SSMs for long-sequence learning.

University of Illinois Urbana-Champaign (UIUC), Research Assistant — *Illinois, U.S. (remote)*
Advisor: Prof. Yu-Xiong Wang — Oct 2023 – Jan 2024

- Contributed to a unified diffusion-based framework that handles multi-modal data generation and dense visual perception together; ran experiments and curated training data.

National University of Singapore (NUS), Research Assistant — *Singapore*
Advisor: Prof. Wei Tsang Ooi — Aug 2023 – Nov 2023

- Explored Neural Radiance Fields for volumetric video representation, integrating NeRF fine-tuning with the LC-checkpoint algorithm to improve compression for streaming systems.

Tsinghua University (THU), Undergraduate Researcher (SRT Program) — *Beijing, China*
Advisor: Prof. He Tian — May 2022 – Sept 2022

- Graphene artificial throat: took part in device fabrication and vibration-signal acquisition, and built machine-learning models for speech pattern recognition from the vibration signals; project rated Grade A.

Selected Awards

Special Excellence Award, Dongguan Science Promotion Association	2025
Outstanding Undergraduate Thesis Award, Tsinghua Univ. (top 5%)	2025
Award for Outstanding Scientific and Technological Innovation, Tsinghua Univ.	2024
3rd iStar Program, Tsinghua Univ. (top 11 teams university-wide)	2024
Third Prize, China-U.S. Young Maker Competition (Beijing Division)	2024
Super Dream Award, Dongguan Science Promotion Association	2023
Zheng Gang Overseas Study Scholarship (4/250), Dept. EE, Tsinghua Univ.	2023
Tsinghua Univ. Awards for Social Work, Volunteer Service, and Fine Arts Excellence	2021 – 2022

Teaching

CS5491: Artificial Intelligence, Teaching Assistant — *Dept. CS, CityU*
Instructor: Prof. Zhichao Lu — Fall 2026

- Ran tutorials and lab sessions on AI concepts and Python implementations; graded assignments and exams.

40231212: Design and Practice of Intelligent Robots, Teaching Assistant — *Dept. EE, THU*
Instructors: Prof. Miling Zhang and Prof. Yueqi Duan — Fall 2024, Spring 2025

- Designed the experiments and teaching materials for the "Path Planning" module; guided students and assessed lab work.

Academic Service

Reviewer:

- Journals: IEEE TMC; IEEE TCDS
- Conferences: NeurIPS 2026/2027; ICML 2026; ICLR 2025/2026/2027; AAAI 2027; IJCAI 2027; AISTATS 2027; ACM MM 2025; CPAL 2026/2027
- Workshops: ODI @ NeurIPS 2026; Trust4RSI-Agent @ AAAI 2027

Student Leadership

Deputy Team Leader, Organization Group, Dept. EE, Tsinghua Univ.	Sept 2022 – Jul 2023
Member, Tsinghua University Student Union	Sept 2021 – Jul 2022
Secretary, Youth League Branch (Best of Show Award, Dept. EE, 2021)	Sept 2020 – Jul 2021

Skills

Programming: Python, PyTorch, R, C/C++, MATLAB, Verilog, LaTeX

Languages: Chinese (native), English (fluent)

Miscellaneous: Badminton (ranked 1st department-wide in mixed doubles)